

Fresh Forecast

Prepared-food waste is a forecasting problem, and forecasting has become cheap. This proposal asks for one pilot store, read-only data access and a modest number of hours to replace gut-feel par sheets with demand forecasts and to measure the result.

Prepared by [name], [role], [store] **For** district and division leadership, prepared foods

Status concept validated in simulation; no company data used **Date** September 2026

Executive summary

Bakery, pizza, hot foods and fresh cases are produced against par sheets that cannot see weather, holidays or trend. The padding those pars carry to avoid empty cases becomes waste; industry-typical waste in these departments runs 15–25% of production. A demand forecast can see every one of those drivers a day ahead.

A working forecasting system already exists. It was built on the author's own time on a fully synthetic store, so that no data-permission or IP question had to be answered before the method was proven. In a full simulated year it cut prepared-food waste by 37% with product availability held at today's level, and by 60% at the profit-optimal setting, closing 81% of the gap between today's practice and perfect knowledge of demand.

The decision requested is a 90-day pilot at one store with a stop gate at week 6, before anything in the kitchen changes. The pilot promises at least a 10% reduction in prepared-food waste dollars against a measured baseline with no material drop in availability. Its downside is a few pallets of mis-produced food; its upside, if it works, is a template for every store in the division.

–37%

waste in the simulated pilot year
with product availability held at
today's level

81%

of the improvement that perfect
knowledge of demand would allow

10–15%

the reduction this proposal actually
promises for a real 90-day pilot

Simulation figures come from a synthetic three-year store, held-out year 2025, reproducible from the project repository. They prove the method; the real dollar figure comes from the pilot's measured baseline.

The decision requested

1. **Approve one pilot store** and name a sponsor at district or division level who owns the result.
2. **Grant written, read-only data access** to daily sales and production counts for 5–10 pilot items at that store, for the duration of the pilot, revocable at any time.
3. **Fund 40–80 paid, non-floor hours** to connect the store's export, run shadow mode and operate the live phase.
4. **Agree a recognition structure in writing**, tied to the measured result and settled before the live phase, so that a successful pilot has somewhere to scale.

The problem is two-sided, which is why it persists

Make too much and we eat the cost of goods. Make too little and we lose the margin and disappoint the customer standing at an empty hot bar at 6 pm. A par sheet solves this with a flat safety pad, but demand is not flat. In the simulated store the demand index runs from 0.60 on a snowy Friday to 1.22 on a sunny one, hot foods swing nearly 40 points between July and January, and Super Bowl Sunday triples whole-pizza demand. A trailing average cannot see any of that coming. A forecast can, because all of those drivers are known the day before.

The reason this has persisted is that the two errors are charged to different ledgers. Waste shows up in shrink; a missed sale shows up nowhere. So the safe thing for a kitchen to do has always been to pad, and nobody has had a tool that priced the two errors against each other item by item.

What already exists

A waste-measurement tool (Phase 1, deployed)

A phone-based logger that records every discarded item with its retail value and produces a store's real baseline in two weeks of use. It is a digital clipboard for what a person observes being thrown away; it connects to nothing.

A working forecasting system (proof of concept, complete)

A demand model of the family used industrially for retail demand forecasting, forecasting a demand *range* for every item, plus a decision rule that turns each range into a morning production quantity by balancing the cost of one wasted unit against the cost of one missed sale. Cheap, high-margin items pad; expensive, low-margin items run lean, automatically. The output is a printed sheet in the kitchen each morning. No integration is required for a pilot: the inputs are a sales export, a public weather forecast and a calendar.

The real-data plumbing (built and rehearsed, unused)

An importer that turns a store's raw export into the model's input, a validation gate, training that takes about a minute per store, and the daily shadow-mode loop. All of it has been run end to end on a mock export built from the synthetic store with realistic dirt added. None of it has seen real data, and no permission has been sought yet: that ordering is deliberate.

Proof of concept: a year replayed

The model trained on the first two simulated years, seeing only what a real store sees, then ran in shadow against the untouched third year. For every day and item it recommended a quantity; the simulation's true demand, which no model ever sees, settled what would have sold, been wasted and been missed. Two benchmark forecasters ran alongside so the network had to earn its seat, and an oracle with perfect knowledge marked the best any method could do.

Held-out year, one store	Waste (retail value)	Waste, % of production	Availability (fill rate)	Total economic cost
Status quo (par sheet)	\$138,462	18.8%	98.1%	\$56,011
Naive trailing average	\$67,278	10.5%	94.6%	\$48,026
Linear model	\$43,617	7.2%	92.8%	\$47,617
Forecast, availability held	\$87,351 (−37%)	12.8%	97.6%	\$41,078
Forecast, profit-optimal	\$55,883 (−60%)	8.8%	95.5%	\$39,723
Oracle (perfect knowledge)	\$57,435	8.9%	96.6%	\$35,788

Economic cost is the total cost of being wrong: wasted units at cost of goods plus missed sales at lost margin. Forecast accuracy: 13.0% weighted error for the network against 15.5% for the trailing average and 15.1% for the linear model. The two benchmarks cut waste by running lean and paying for it in missed sales, which is why their economic cost barely moves.

The two forecast rows are the same model with a different availability target. **Availability held** is the pilot setting: \$51,111 less food thrown away per year at retail value, with customers seeing an empty case about as often as they do today. **Profit-optimal** is where the decision rule would go on its own; the availability it gives up is a business choice to be made item by item, not a model setting. Savings accrue steadily through the year, about \$300 a week per store in economic cost from the first week of live operation.

What the simulation proves, and what it does not

It proves the method: demand with realistic weekday, weather, holiday and seasonal structure is learnable from sales data alone, censored by sellouts as a real store's data is, and production driven by those forecasts beats a well-run par sheet by a wide margin. It does not prove this division's dollar figure. The simulated store is a plausible invention, and its simulated manager, though deliberately competent, is an invention too. That is why this proposal promises 10–15% in a real pilot rather than 37–60%, and why the pilot's measured baseline, not the simulation, is the number it is judged against. No company data of any kind was used to build any of this.

The 90-day pilot

WEEKS 1-2

Baseline

Log every discarded prepared-food item across 5–10 pilot items with the Phase 1 tool.

Output: this store's real waste number, in dollars per year.

WEEKS 3-6

Shadow mode

With data access granted, the model retrains on this store's history and prints its morning sheet. The kitchen ignores it; we record what it would have done.

Output: forecast accuracy against reality, scored against five go/no-go criteria fixed in advance.

WEEKS 7-13

Live

The kitchen follows the sheet. The manager can override any line on any day; overrides are logged, not fought.

Output: measured waste and sellout change versus the baseline.

Success criterion, agreed in advance: at least a 10% reduction in prepared-food waste dollars versus the Phase 1 baseline, with no material drop in on-shelf availability. If shadow mode shows the forecasts are not accurate enough for this store, the pilot stops at week 6 and nothing in the kitchen has changed.

Governance and cost

- **Sponsor.** One district or division leader owns the pilot, receives the weekly shadow-mode report, and makes the week-6 go/no-go call against the pre-agreed criteria.
- **Data.** Read-only, scoped to the pilot items at one store, for the pilot's duration, revocable at any time. Nothing leaves company systems except the printed sheet. The request is made before any real data is touched.
- **Cost.** 40–80 paid, non-floor hours, most of them in weeks 3–6. No software licences, no integration work, no hardware beyond a printer the kitchen already has.
- **Downside.** Bounded to the live phase: a few pallets of mis-produced food over six weeks, at a store whose manager can revert any item to its par on any day.
- **IP and recognition.** Nothing was built on company data, so ownership can be settled cleanly on the way in. The recognition structure is agreed in writing before the live phase.

If it works

One model serves every item, so the pilot is also the template. The same architecture scales from nine items at one store to every prepared-foods case in a district without redesign; retraining per store takes about a minute, and the morning sheet is the only interface a kitchen ever sees. The table below is arithmetic on the simulated store, not a forecast of Harris Teeter's numbers. Its job is to show which column the pilot replaces with a measured fact.

Per year	Food not wasted (retail, simulated)	Economic cost avoided (simulated)	Promise applied (10- 15% of waste)
1 store	\$51,111	\$14,933	\$13,846 – \$20,769
20 stores	\$1,022,216	\$298,664	\$276,924 – \$415,386
50 stores	\$2,555,539	\$746,660	\$692,310 – \$1,038,466
100 stores	\$5,111,078	\$1,493,320	\$1,384,621 – \$2,076,931

Simulated store: \$138,462 of prepared-food waste at retail per year under the par sheet; availability-held forecast setting. Replace the first row with the pilot's measured baseline and the rest follows. Do not read the larger rows as projections; read them as the reason a single-store pilot is worth 80 hours.

Risks, named

Risk	Mitigation
Forecasts do not transfer from simulation to this store	Shadow mode is the gate: four weeks of predictions scored against reality before production changes at all.
Availability suffers and customers notice	The pilot runs at the availability-held setting; sellouts are tracked daily and any item can revert to its par instantly.
The kitchen does not trust the sheet	Manager override is a feature, logged not fought; the pilot measures the blend, which is how it would really run.
The store's export is dirtier than expected	The importer has been rehearsed against a deliberately dirty mock export. The first real one will still surprise us, and the hours ask covers that.
Data access or IP questions	Nothing was built on company data; access is scoped, read-only, written and revocable. IP and recognition are settled in writing before the live phase.

Method notes. Demand model: a global quantile-regression neural network (GRU encoder over each item's trailing 28 days, learned item embeddings, calendar, weather and holiday covariates; pinball loss across 11 quantiles, with sellout days treated as censored since sales only bound demand from below when the case sold out). Decision layer: per-item newsvendor critical fractile, rounded to batch sizes; the availability-held variant raises each item's target until the year's fill rate matches the par sheet's. Validation: trained on simulated 2023–24, tested untouched on 2025 against naive, linear and oracle policies. The synthetic dataset, the trained checkpoint and the results file are frozen and checked in to the project repository, and the backtest reproduces the results byte for byte. An interactive results dashboard and a slide version of this proposal accompany this document.